Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Oxymercuration-demercuration of $CH_3CH = CH_2$ produces

(A) $CH_3CH_2CH_2OH$

(B) $CH_3CH(OH)CH_3$

(C) $CH_3CH(OH)CH_2OH$

(D) CH_3COCH_3

Q.2

Which of the following statements is not correct?

(A) The substitution of hydroxyl group by a halogen group in alcohol is an electrophilic substitution reaction.

(B) Alcohols are weak acids as well as weak bases.

(C) A secondary alcohol on oxidation gives a carboxylic acid containing the lesser number of carbon atoms.

(D) A primary alcohol on oxidation gives a carboxylic acid containing the same number of carbon atoms.

Q.3

The order of reactivity of hydrogen halides with an alcohol is

(A) $HCl > HBr > HI$

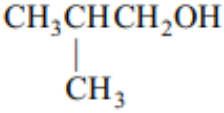
(B) $HCl < HBr < HI$

(C) $HCl > HBr < HI$

(D) $HCl < HBr > HI$

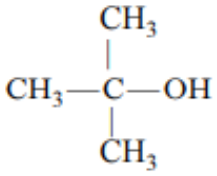
Q.4

The compound which gives the most stable carbonium ion on dehydration is

(A) 

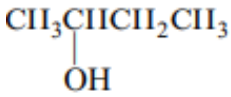
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(B)



(C) $CH_3CH_2CH_2CH_2OH$

(D)



Q.5

The product obtained when excess of HI reacts with glycerol is

(A) 1, 2, 3-Triodopropane

(B) Allyl iodide

(C) 1, 2-Diiodopropane

(D) Isopropyl iodide

Q.6

The relative order of basicity of conjugate bases is

(A) $OH^- < OR^- < HC \equiv C^-$

(B) $OH^- < HC \equiv C^- < OR^-$

(C) $HC \equiv C^- < OH^- < OR^-$

(D) $HC \equiv C^- < OR^- < OH^-$

Q.7

Which of the following diols would cleave into two fragments with HIO_4

(A) $1,3-$hexanediol

(B) $2,4-$hexanediol

(C) $1,6-$hexanediol

(D) $3,4-$hexanediol

Q.8

A liquid is mixed with ethanol and a drop of concentrated H_2SO_4 is added. A compound with a fruity smell is formed. The fruity smelling compound formed is:

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(A) CH_3COCH_3

(B) $CH_3COOC_2H_5$

(C) CH_3OH

(D) HCHO

Q.9

In the hydroboration - oxidation reaction of propene with diborane, H_2O_2 and $NaOH$, the organic compound formed is:

(A) CH_3CH_2OH

(B) $CH_3CHOHCH_3$

(C) $CH_3CH_2CH_2OH$

(D) $(CH_3)_3\,COH$

Q.10

Which of the following orders is true regarding the acidic nature of phenol?

(A) Phenol $>$ *o*-Cresol $>$ *o*-Nitrophenol

(B) Phenol $>$ *o*-Cresol $<$ *o*-Nitrophenol

(C) Phenol $<$ *o*-Cresol $>$ *o*-Nitrophenol

(D) Phenol $<$ *o*-Cresol $<$ *o*-Nitrophenol

Q.11

Which of the following statements is not correct?

(A) Picric acid is $2, 4, 6$-trinitrophenol.

(B) Kolbe reaction helps in converting phenol to salicyclic acid.

(C) Reimer-Tiemann reaction involving phenol and chloroform converts the former into salicyclic acid.

(D) Phenol gives characteristic colour with $FeCl_3$. This test is due to enol structure.

Q.12

Phenol on treating with concentrated H_2SO_4 at $15 - 20^\circ C$ mainly produces

(A) phenol-2-sulphonic acid

(B) phenol-4-sulphonic acid

(C) a 50% mixture of ortho and para phenol sulphonic acid

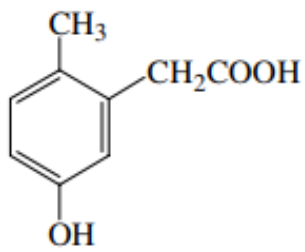
(D) phenol-2, 4, 6-trisulphonic acid

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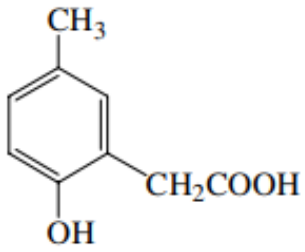
Q.13

p-cresol reacts with chloroform in alkaline medium to give the compound A which adds hydrogen cyanide to form the compound B. The latter on acidic hydrolysis gives chiral carboxylic acid. The structure of the carboxylic acid is

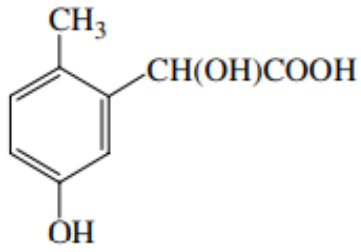
(A)



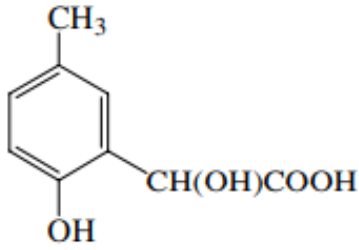
(B)



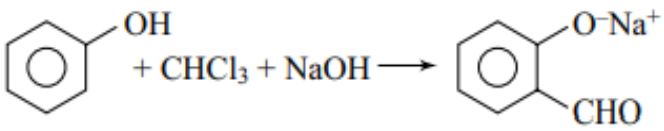
(C)



(D)



Q.14



The electrophile involved in the above reaction is

(A) formyl cation ($^{\oplus}(CHO)$

(B) dichloromethyl cation $(CHCl_2)$

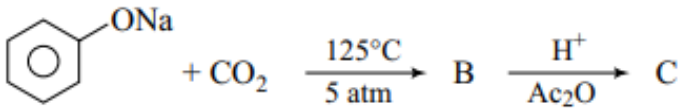
(C) dichlorocarbene $(: CCl_2)$

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(D) trichloromethyl anion $\left(\overset{+}{\mathrm{C}}\mathrm{Cl}_3\right)$

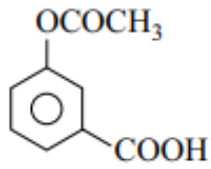
Q.15

Sodium phenoxide when heated with CO_2 under pressure at $125^\circ C$ yields a product which on acetylation, produces C.

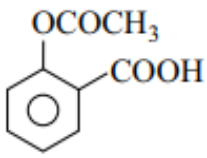


The major product C would be

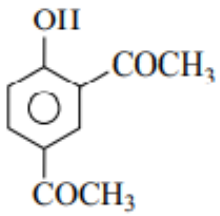
(A)



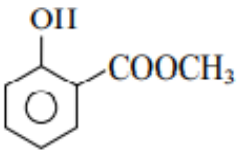
(B)



(C)



(D)



Q.16

The alkoxy group in benzene is

(A) ortho directing

(B) para directing

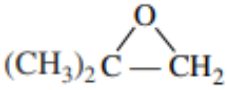
(C) ortho and para directing

(D) meta directing

Q.17

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The reaction of



with CH_3OH in (i) acid H^+, and (ii) base CH_3O^-, respectively, give

(A) $(CH_3)_2\,C\,(OCH_3)\,CH_2OH$ and $(CH_3)_2\,CH(OH)CH_2OCH_3$

(B) $(CH_3)_2\,C\,(OCH_3)\,CH_2OH$ and $(CH_3)_2\,C\,(OCH_3)\,CH_2OH$

(C) $(CH_3)_2\,C\,(OCH_3)\,CH_2OCH_3$ and $(CH_3)_2\,C(OH)CH_2OH$

(D) $(CH_3)_2\,C(OH)CH_2OH$ and $(CH_3)_2\,C\,(OCH_3)\,CH_2OCH_3$

Q.18

Allyl phenyl ether can be prepared by heating:

(A) $C_6H_5Br + CH_2 = CH - CH_2 - ONa$

(B) $CH_2 = CH - CH_2 - Br + C_6H_5ONa$

(C) $C_6H_5 - CH = CH - Br + CH_3 - ONa$

(D) $CH_2 = CH - Br + C_6H_5 - CH_2 - ONa$

Q.19

A molecule consumes 2 molecules of HIO_4. The number of carbon-carbon bond broken by this acid is

(A) one

(B) two

(C) three

(D) none

Q.20

The major product obtained on interaction of phenol with sodium hydroxide and carbon dioxide is

(A) Salicylic acid

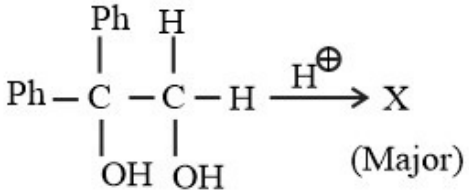
(B) Phthalic acid

(C) Benzoic acid

(D) Salicyladehyde

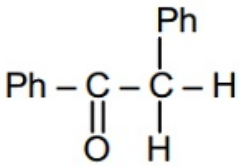
Q.21

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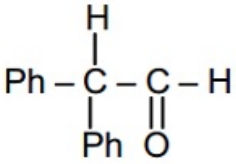


Identify X.

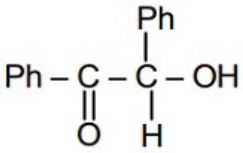
(A)



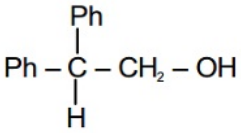
(B)



(C)



(D)



Q.22

Which reagent can convert acetic acid into ethanol?

(A) Na + alcohol

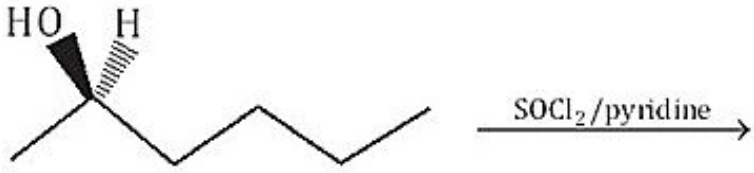
(B) $\mathrm{LiAlH_4}$ + ether

(C) $\mathrm{H_2} + \mathrm{Pt}$

(D) $\mathrm{Sn} + \mathrm{HCl}$

Q.23

Give the major product of the following reaction < br /> < br />

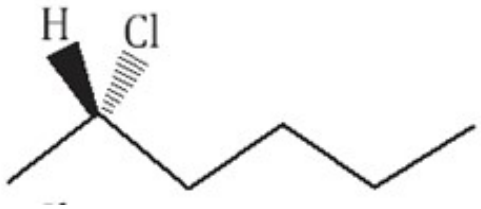


(A)



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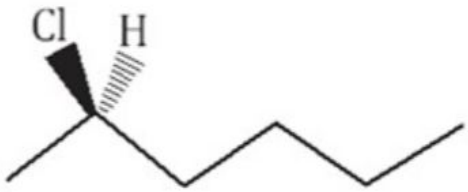
(B)



(C)

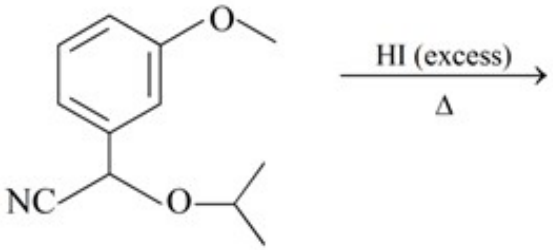


(D)

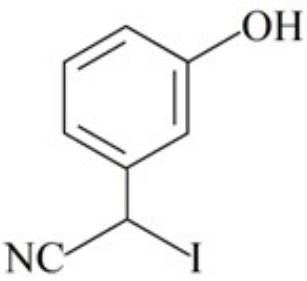


Q.24

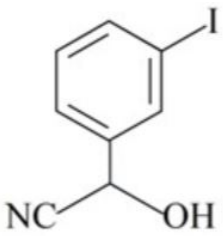
The major product of the following reaction is : < br /> < br />



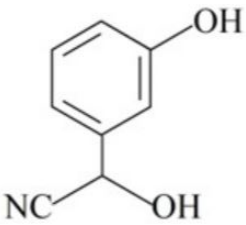
(A)



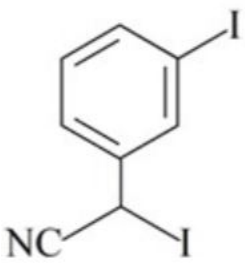
(B)



(C)



(D)



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Q.25

Formaldehyde gives an additive product with methyl magnesium iodide which on aqueous hydrolysis gives -

(A) Isopropyl alcohol

(B) Ethyl alcohol

(C) Methyl alcohol

(D) Propyl alcohol

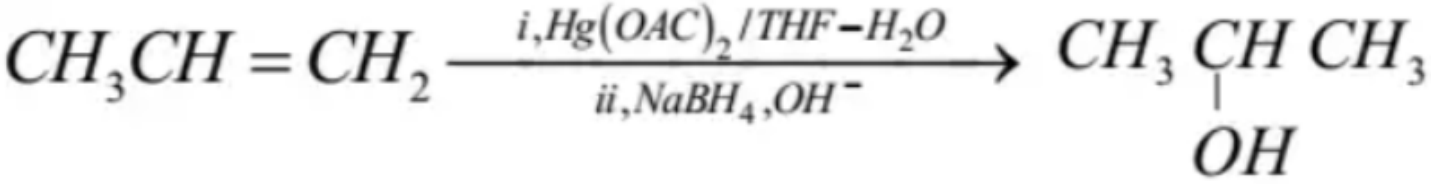
Answers & Solutions

Q.1 Answer:

$CH_3CH(OH)CH_3$

Solution:

In oxymercuration - demercuration, more stable carbocation is formed and water attacks on more substituted carbon. The reaction is as follows:



Q.2 Answer:

The substitution of hydroxyl group by a halogen group in alcohol is an electrophilic substitution reaction.

Solution:

Explanation of options:

(a) Replacement of $-OH$ by a halogen in an alcohol is nucleophilic substitution reaction. It is the protonated alcohol which acts as a substrate.

(b) Alcohols are acidic enough to react with active metals to liberate hydrogen gas. They are basic enough to accept a proton from strong acids.

(c) It gives a ketone containing the same number of carbon atoms. Further oxidation will give carboxylic acid containing lesser number of carbon atoms.

(d) A primary alcohol, for example, methanol (1 C atom), on oxidation gives a carboxylic acid containing the same number of carbon atoms, that is formic acid (1 C atom).

So, the correct option is (a) since the statement is wrong

Q.3 Answer:

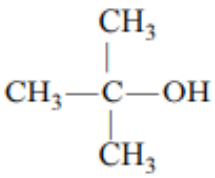
$HCl < HBr < HI$

Solution:

The order of reactivity is $HCl < HBr < HI$.

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Q.4 Answer:



Solution:

Order of stability of carbocations is

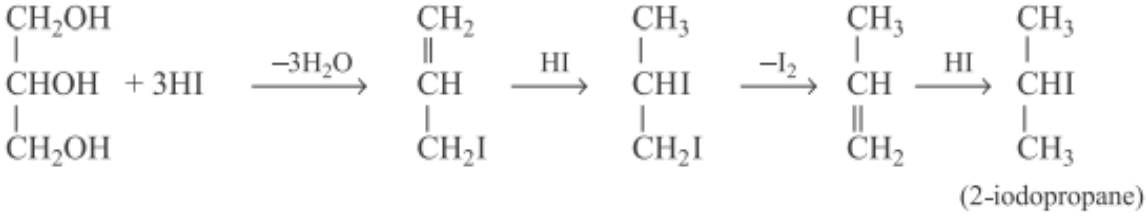
primary $<$ secondary $<$ tertiary

Dehydration of primary alcohol gives primary carbocation, and dehydration of secondary, tertiary alcohol gives secondary and tertiary carbocation respectively.

Q.5 Answer:

Isopropyl iodide

Solution:



Q.6 Answer:

$OH^- < OR^- < HC \equiv C^-$

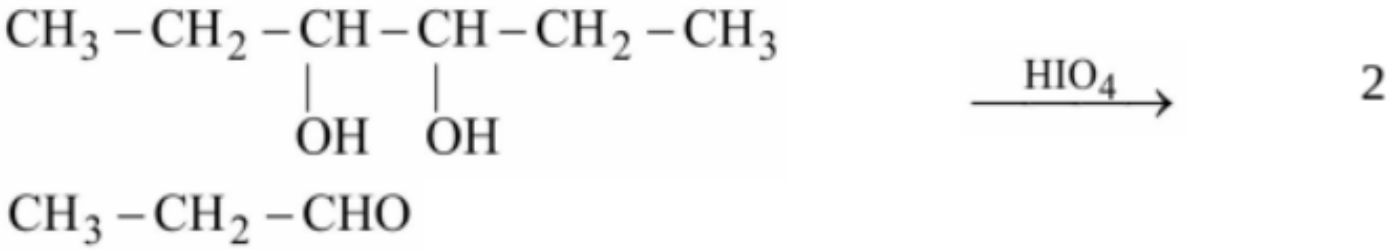
Solution:

The relative order of basicity of conjugate base is opposite to that of acid.

Q.7 Answer:

$3, 4-$hexanediol

Solution:



Q.8 Answer:

$CH_3COOC_2H_5$

Solution:

The fruity smell is evolved due to the formation of ester when ethanol and acid are treated in the presence of concentrated H_2SO_4.The reaction is

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$$CH_3COOH + HOC_2H_5 \overset{H^+}{\rightleftharpoons} \underset{\text{Ester}}{CH_3COOC_2H_5} + H_2O$$
(fruity smell)

Q.9 Answer:

$CH_3CH_2CH_2OH$

Solution:

In the hydroboration - oxidation reaction, an alkene reacts with BH_3 to give alkyl boranes which on oxidation with alkaline H_2O_2 gives alcohol. The product obtained in the accordance with the anti-Markovnikov's rule.\n$CH_3 - CH = CH_2 + BH_3 \rightarrow CH_3CH_2CH_2BH_2 \xrightarrow{H_2O_2,OH^-} CH_3CH_2CH_2OH$

Q.10 Answer:

Phenol $>$ *o*-Cresol $<$ *o*-Nitrophenol

Solution:

Methyl group is activating group (electron release to benzene) and nitro group is deactivating group (electron withdrawal group). Methyl group will make release of H^+ more difficult and nitro group will make it more easy.

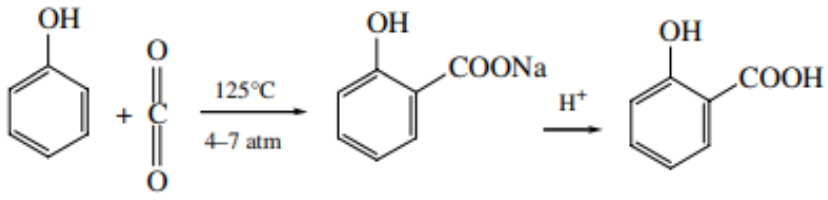
Q.11 Answer:

Reimer-Tiemann reaction involving phenol and chloroform converts the former into salicyclic acid.

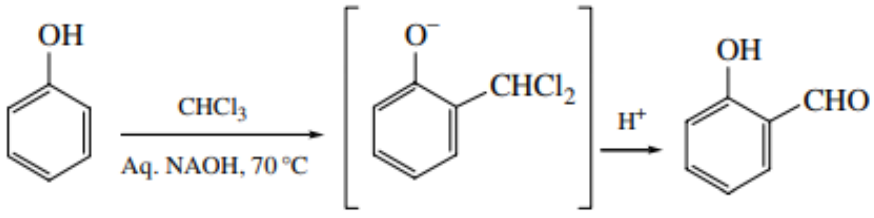
Solution:

Explaining the options

(b) The reactions are



(c) The compound formed is salicylaldehyde.



Q.12 Answer:

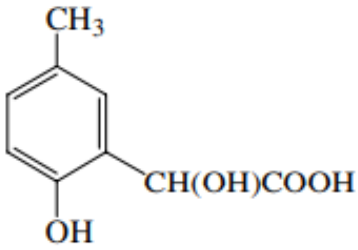
phenol-2-sulphonic acid

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Solution:

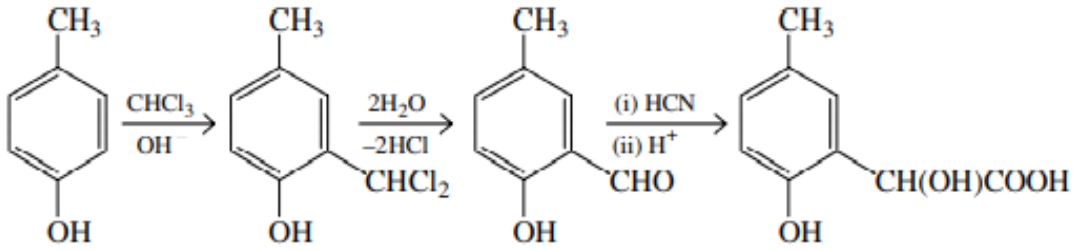
At low temperature, ortho-product is obtained.

Q.13 Answer:



Solution:

The main substituent which decides the position of electrohilic substitution in *p*-cresol is $-$ OH group. Therefore, the electrophilic substitution occurs at ortho position to the $-$OH group. The reactions involved are as follows:

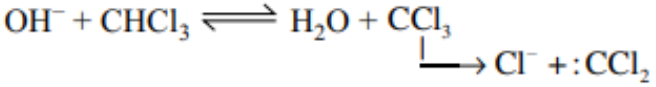


Q.14 Answer:

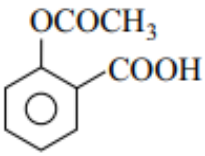
dichlorocarbene $(:CCl_2)$

Solution:

The electrophile is :CCl $_2$ generated by the rection:

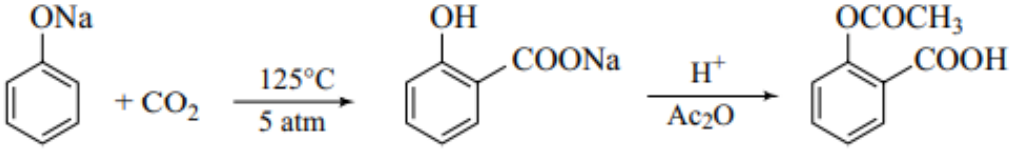


Q.15 Answer:



Solution:

The given reaction is Kolbe reaction followed by acetylation of —OH group.



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Q.16 Answer:

ortho and para directing

Solution:

Alkoxy (R-O) group with +M -effect activates the ortho and para positions in the ring towards electrophilic substitution

Q.17 Answer:

$(CH_3)_2\,C\,(OCH_3)\,CH_2OH$ and $(CH_3)_2\,CH(OH)CH_2OCH_3$

Solution:

There occurs cleavage of ether linkage in oxiranes (cyclic three membered ring including oxygen). In acidcatalyzed reaction, nucleophile $^-OCH_3$ from CH_3OH attaches to the more substituted carbon where as in basecatalyzed reaction, it is attached to the less substituted carbon.

Q.18 Answer:

$CH_2 = CH - CH_2 - Br + C_6H_5ONa$

Solution:

To prepare an ether containing a phenyl group, the only choice is to take sodium phenoxide and not phenyl bromide as the latter is much less reactive towards nucleophilic substitution reaction.

Q.19 Answer:

two

Solution:

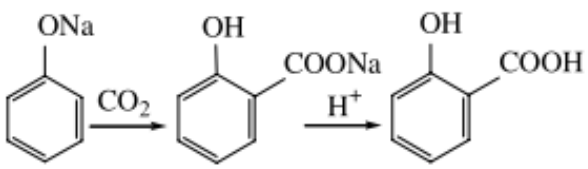
The number of $C - C$ bond broken is equal to the number of HIO_4 molecules used.

Q.20 Answer:

Salicylic acid

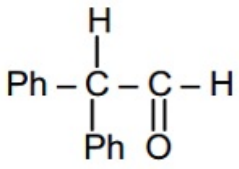
Solution:

The main reaction is



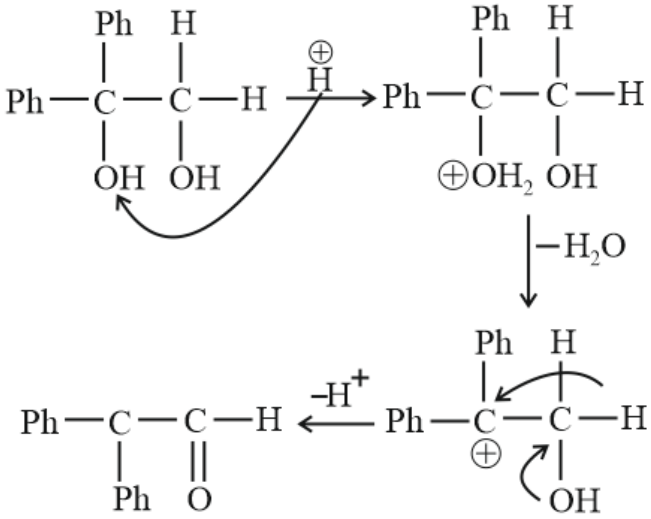
Salicyclic acid

Q.21 Answer:



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Solution:



Q.22 Answer:

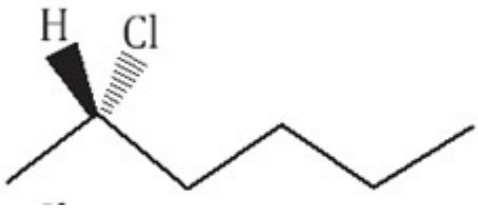
$LiAlH_4$ + ether

Solution:

$LiAlH_4$ + **ether, is reducing agent.**

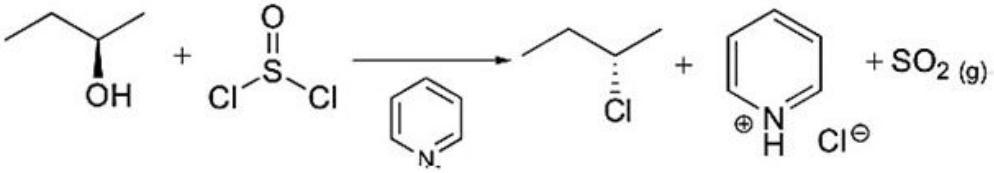
It reduce all types of carboxylic acids to alcohols. So, this reagent can be use to convert acetic acid into ethanol. $CH_3COOH \xrightarrow{LiAlH_4/ether} CH_3CH_2OH$

Q.23 Answer:

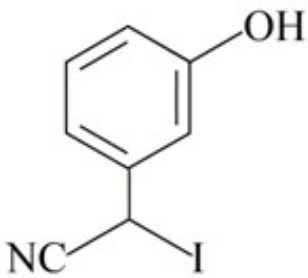


Solution:

Conversion of alcohols to chlorides with $SOCl_2$ proceeds with inversion

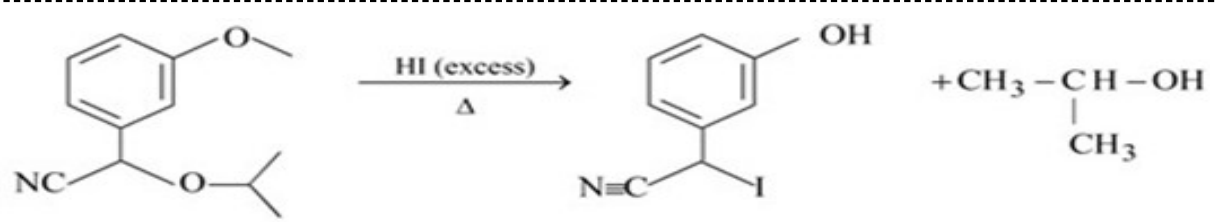


Q.24 Answer:



Solution:

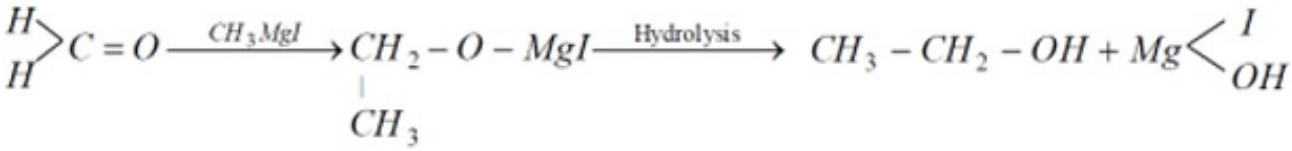
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Q.25 Answer:

Ethyl alcohol

Solution:



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